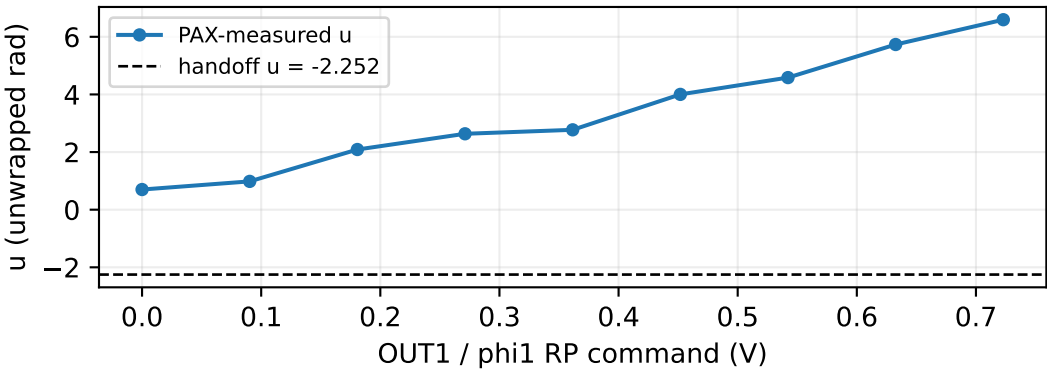
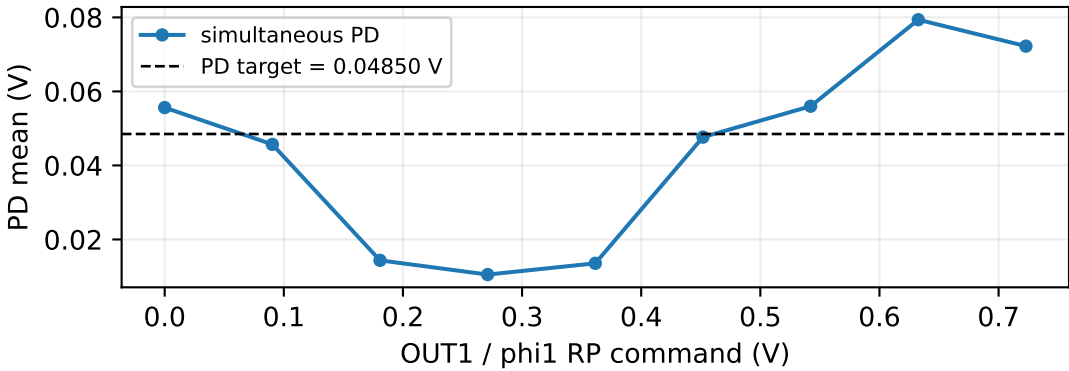


Phi1 PD FPGA lock — 021422_phi1-pd-hybrid-test_autonomous-hybrid-gentle-confirm-0

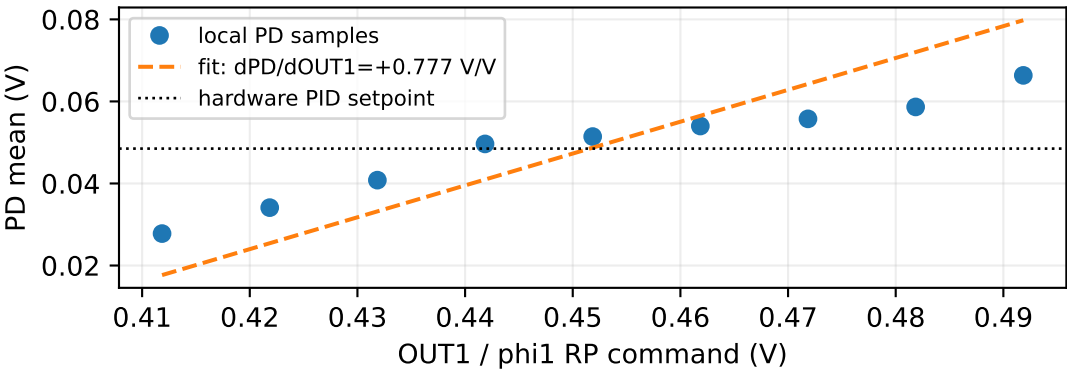
PAX-assisted ϕ_1 rough map



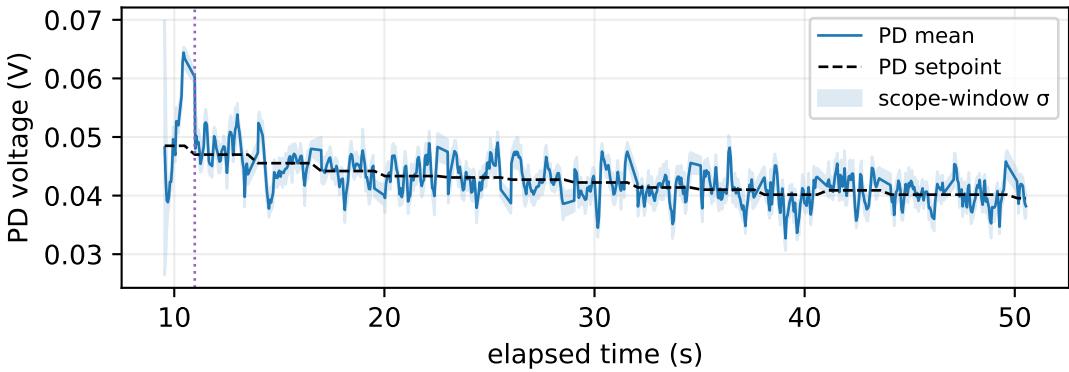
PD value associated with rough u map



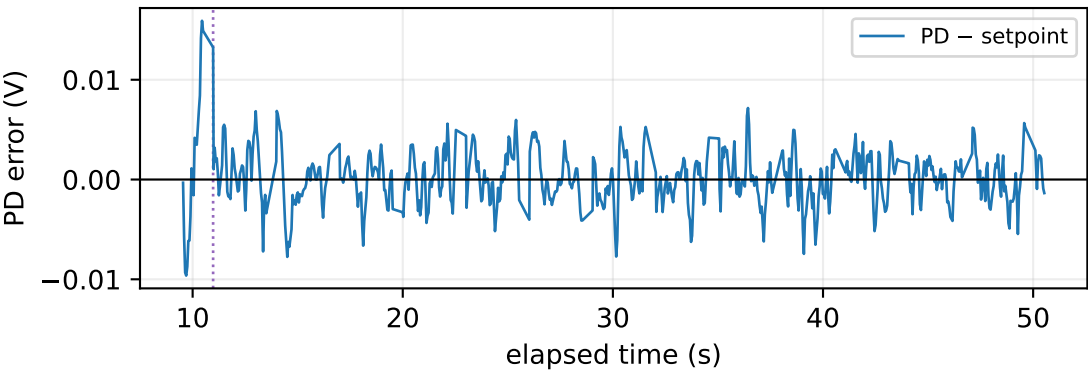
Automatic local PD calibration



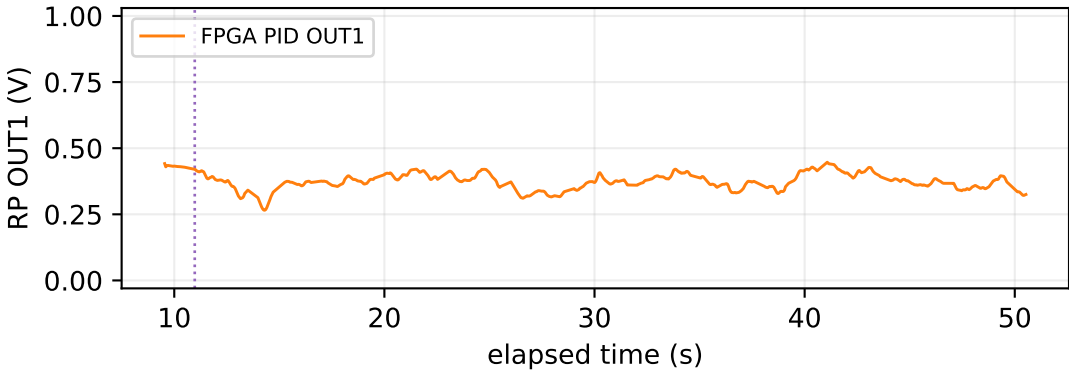
FPGA PD lock



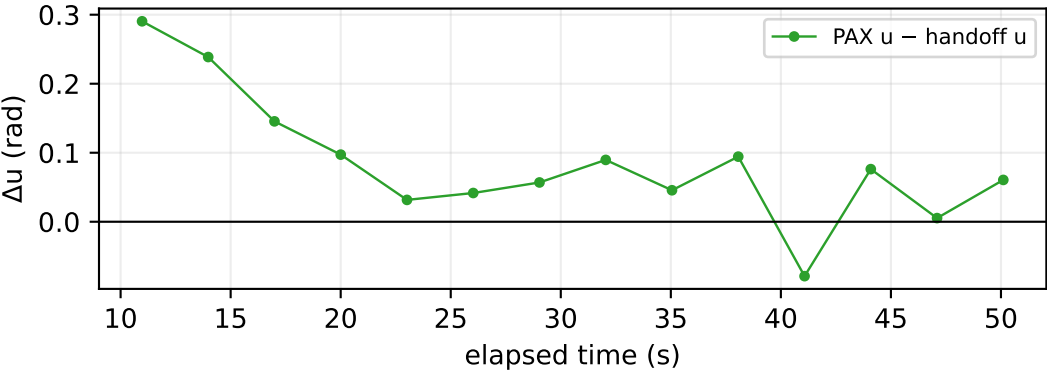
PD locking error



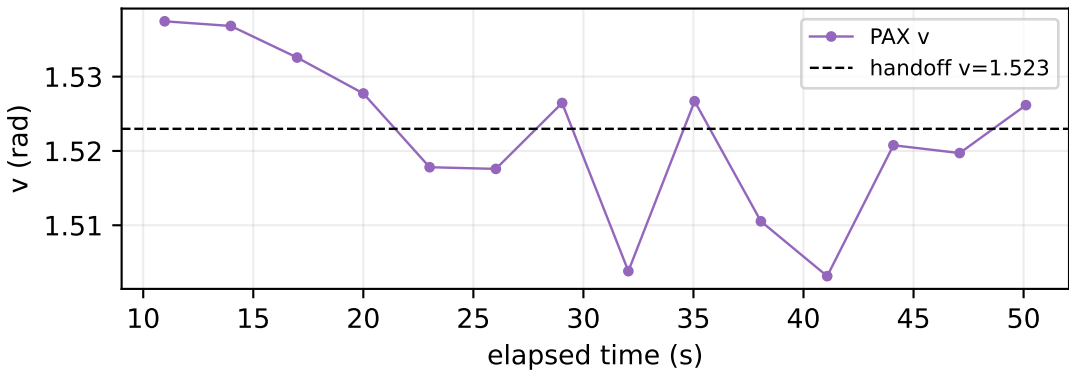
Continuous FPGA actuator output



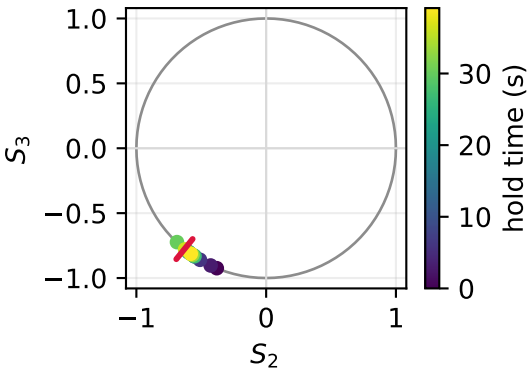
Independent PAX azimuth validation



PAX polar-angle cross-check



Held lock trajectory on the Poincare equator



PAX handoff target $u = -2.2517$ rad | initial PD setpoint = 0.04850 V | local slope = +0.7765 V/V
FPGA PID: P = -0.283, I = -5 Hz | duration = 41.0 s | median |PD error| = 0.00163 V (late 0.00156 V)
OUT1 = 0.265...0.447 V | PD setpoint = 0.03955...0.04850 V | PAX median $|\Delta u|$ = 0.078 rad (14 snapshots)